

Dihybrid Cross & Independent Assortment

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Dihybrid Cross & Independent Assortment

15 questions · answer key on the last page

- Q1.** A dihybrid cross involves parents differing in how many pairs of contrasting characters?
- (a) One
 - (b) Two
 - (c) Three
 - (d) Four
- Q2.** In Mendel's dihybrid cross (RRYY x rryy), the F1 generation phenotype is:
- (a) Round, Green
 - (b) Wrinkled, Yellow
 - (c) Round, Yellow
 - (d) Wrinkled, Green
- Q3.** The classic F2 phenotypic ratio obtained in a dihybrid cross is:
- (a) 3:1
 - (b) 1:2:1
 - (c) 9:3:3:1
 - (d) 1:1:1:1
- Q4.** In the dihybrid cross for seed shape and seed colour, which traits are dominant?
- (a) Wrinkled and Green
 - (b) Round and Yellow
 - (c) Round and Green
 - (d) Wrinkled and Yellow
- Q5.** A heterozygous plant RrYy produces how many types of gametes?
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 8
- Q6.** Mendel's Law of Independent Assortment is based on which type of cross?
- (a) Monohybrid cross
 - (b) Dihybrid cross
 - (c) Test cross
 - (d) Reciprocal cross
- Q7.** Mendel's work was rediscovered in 1900 by three scientists working independently. Which of the following was NOT one of them?
- (a) Hugo de Vries
 - (b) Carl Correns
 - (c) Von Tschermak
 - (d) Thomas Hunt Morgan
- Q8.** Mendel published his findings in the proceedings of which society?
- (a) Royal Society of London
 - (b) Natural History Society of Brunn
 - (c) National Academy of Sciences
 - (d) Linnean Society
- Q9.** In the F2 of a dihybrid cross, the ratio of Round-Yellow : Round-Green : Wrinkled-Yellow : Wrinkled-Green is:
- (a) 9:3:3:1

- (b) 3:3:9:1
- (c) 1:3:3:9
- (d) 9:1:3:3

Q10. The 9:3:3:1 dihybrid ratio arises from combining which two monohybrid ratios?

- (a) (1:2:1) x (1:2:1)
- (b) (3:1) x (3:1)
- (c) (1:1) x (1:1)
- (d) (3:1) x (1:1)

Q11. In the F₂ of a dihybrid cross with 16 total combinations, how many individuals show entirely new (recombinant) combinations not seen in either parent?

- (a) 2
- (b) 4
- (c) 14
- (d) 16

Q12. In the F₂ generation of RrYy x RrYy, the genotype "RRYY" represents:

- (a) A new recombinant type
- (b) The pure dominant parental type
- (c) The pure recessive parental type
- (d) A heterozygous type

Q13. In the F₂ generation of RrYy x RrYy, the genotype "rryy" represents:

- (a) A recombinant type
- (b) The pure dominant parental type
- (c) The pure recessive parental type
- (d) A heterozygous type

Q14. The full F₂ genotypic ratio of a dihybrid cross (RrYy x RrYy) is:

- (a) 1:2:1
- (b) 9:3:3:1
- (c) 1:2:1:2:4:2:1:2:1
- (d) 1:1:1:1

Q15. Why was Mendel's work not recognized until 1900?

- (a) He never published his results
- (b) Lack of communication/publicity, his 'discrete factor' idea was not readily accepted, and his mathematical approach was unfamiliar
- (c) His pea plants were the wrong species
- (d) He worked in a country with no scientific societies

Answer Key & Explanations

- Q1.** (b) Two — A dihybrid cross is between two homozygous individuals differing in two pairs of contrasting characters, such as seed shape and seed colour.
- Q2.** (c) Round, Yellow — Since Round and Yellow are both dominant, the F1 (RrYy) from RRYy x rryy shows the round, yellow phenotype.
- Q3.** (c) 9:3:3:1 — Selfing a dihybrid F1 (RrYy x RrYy) gives the characteristic F2 phenotypic ratio of 9:3:3:1.
- Q4.** (b) Round and Yellow — Round seed shape and yellow seed colour are the dominant traits over wrinkled and green respectively.
- Q5.** (c) 4 — A RrYy plant produces four types of gametes in equal proportion: RY, Ry, rY, and ry.
- Q6.** (b) Dihybrid cross — Unlike the Laws of Dominance and Segregation, the Law of Independent Assortment is based on Mendel's dihybrid cross experiments.
- Q7.** (d) Thomas Hunt Morgan — Hugo de Vries, Carl Correns, and Von Tschermak rediscovered Mendel's work in 1900; Thomas Hunt Morgan later worked on *Drosophila* and sex linkage.
- Q8.** (b) Natural History Society of Brunn — Mendel published his results in the 'Proceedings of the Natural History Society' of Brunn in 1865.
- Q9.** (a) 9:3:3:1 — The dihybrid F2 phenotypic ratio is 9 Round-Yellow : 3 Round-Green : 3 Wrinkled-Yellow : 1 Wrinkled-Green.
- Q10.** (b) (3:1) x (3:1) — The 9:3:3:1 phenotypic ratio is the product of two independent monohybrid 3:1 ratios (3 Round:1 Wrinkled) x (3 Yellow:1 Green).
- Q11.** (c) 14 — Of the 16 F2 combinations, only RRYy (pure dominant) and rryy (pure recessive) match the parents; the remaining 14 are new recombinant combinations.
- Q12.** (b) The pure dominant parental type — RRYy is homozygous dominant for both genes and matches the original pure dominant parent (RRYy).
- Q13.** (c) The pure recessive parental type — rryy is homozygous recessive for both genes, matching the original pure recessive parent (rryy).
- Q14.** (c) 1:2:1:2:4:2:1:2:1 — The complete dihybrid F2 genotypic ratio across all nine genotype classes is 1:2:1:2:4:2:1:2:1.
- Q15.** (b) Lack of communication/publicity, his 'discrete factor' idea was not readily accepted, and his mathematical approach was unfamiliar — Mendel's work went unrecognised due to poor publicity, resistance to his discrete-factor concept given continuous variation in nature, and unfamiliarity with his mathematical approach.